

FPV RC Arena — Dry Run Report

Six simulated passes through Build Plan v1 (draft of 11 Sep 2026), each run as if executing it. Every error found was fixed in the plan; this is the record of what changed and why.

Run 1 — Procurement: ordering the parts list

#	Error	Effect	Fix in plan
1.1	Opening mix was 3 monitor + 1 goggles stations	Over-scoped for a first venue; four G29 cockpits before one has earned money	2 stations: 1 monitor + 1 goggles. Extra monitor stations priced as a later add
1.2	Car kits were counted per station (one each)	A station needs 2 cars in rotation (one running, one charging); half the kits were missing	Kits per fitted car: 4 + 1 spare cartridge. Budget rebuilt
1.3	ELRS TX module is a module-bay part — no standalone power/CRSF wiring was specified	Module can't be powered or driven without wiring the bay pins	Bay-pin wiring, 5 V feed via E-stop chain, unique bind phrase per station
1.4	CRSF on the module is one-wire half-duplex; the plan wired a full-duplex USB-UART straight in	Bus contention, no telemetry, likely no link	RXD direct, TXD via 1 k Ω ; CP2102N/FT232RL for 400 k baud; bench-verify
1.5	Marshal master stop used a relay per station	Extra parts for a two-station arena	One 2-NC mushroom in series with both TX 5 V feeds

Run 2 — Bench build of the first station

#	Error	Effect	Fix in plan
2.1	RushFPV Tank Solo needs 7 V minimum; every candidate car is 2S (7.4 V nominal, ~6.5 V under brushed-motor load)	VTX browns out under throttle — video drops exactly when the car is moving	SpeedyBee TX800 (5 V in) fed by a 5 V 3 A UBEC + LC filter. Camera takes 5–36 V, fine
2.2	Hardware failsafe set throttle to "light brake" 1400 μ s	On a brushed ESC, below-neutral after neutral = reverse . Failsafe could drive the car backwards	Failsafe = 1500/1500 neutral + ESC drag brake where offered
2.3	Software brake pulse applied unconditionally	Same reverse risk if the car was stationary or reversing	Brake only if last motion was forward, else neutral

#	Error	Effect	Fix in plan
2.4	No behaviour defined for a Pi crash or hang	TX module may keep sending last channel values; car keeps driving	Heartbeat watchdog: missing-pulse detector drops the TX module relay (reuses the relay dropped in 1.5)
2.5	No "disarmed on reconnect" rule	After an E-stop release the car could resume the last command	Every (re)connect starts disarmed at neutral until staff clears
2.6	G29 Linux mode not stated	Wheel enumerates differently in PS3/PS4 mode; FFB can fight the driver	PS3 mode, light centring spring only

Run 3 — Track and video test

#	Error	Effect	Fix in plan
3.1	Channel plan listed four channels for two cars	Unnecessary; adjacent Raceband channels bleed at close range	R3 + R6 (max separation); R4/R5 reserved
3.2	Goggles ship with linear whips; only the monitor got RHCP antennas	Goggle feed multipath-noisy indoors	2 × RHCP antennas added to the goggles delta
3.3	Gate tested "four video links"	Wrong count	Gate item 7 = both VTX + goggles clean, venue Wi-Fi on, full ESC load

Run 4 — Customer sessions (operations)

#	Error	Effect	Fix in plan
4.1	Goggles station carried its own driver monitor and a second marshal monitor	₹9–11k double-count; the station's 7-inch screen already has its own receiver	Base monitor = attendant's marshal screen. Goggles delta down to ₹19–34.5k
4.2	Session end = instant stop mid-track	Marshal walks the track every session; reset over 3 min	Last 20 s at 20 % "return to pit" mode
4.3	Reset test ignored battery swaps	2S brushed cars last 15–20 min; swap is part of reset	Gate item 11 includes battery swap; 2 cars per station
4.4	No staffing number	Attendant cannot watch a goggles guest and marshal the track	2 on floor: marshal + goggles attendant

Run 5 — Failure drills

#	Error	Effect	Fix in plan
5.1	Gate had no Pi-crash test	Watchdog (2.4) untested	Gate item 4: kill process / pull Pi power → failsafe within 1.5 s, 10/10
5.2	Master stop test covered two links but not the pendant	Attendant's stop untested	Gate item 5 covers station mushroom, pendant, master stop
5.3	"Prototype station" was defined as monitor + goggles on one station	Goggles lane never tested as its own station with its own loom	Monitor station is the prototype; goggles station is built second from the proven loom, then gate items 7 and 12 re-run

Run 6 — Two design changes re-run end to end (11 Sep)

Change A — separate video battery per car. Adopted. The car pack was the source of every video problem found in run 2 (sag below the VTX minimum, brushed-motor noise). A dedicated 2S pack feeding a 5 V UBEC removes both, and the LC filter with them. Cost: about +Rs 1,000–1,500 per car (two small packs + buzzer alarm, minus the filter). Runtime is 1–2 h per pack against 15–20 min per car pack, so it is one swap every fourth car-battery swap. Residual: radiated motor noise into the camera cable is small; route the lead away from the motor. Small 2S packs are thin in Indian stock – buy all ten at once.

Change B — 27-inch gaming monitor at the cockpit. Adopted with a gate. The 7-inch unit stays as the receiver + DVR + marshal screen and its AV out feeds a CVBS-to-HDMI upscaler into an LG 27GS60F (Rs 12,289, Computech Store, in stock). Two honest limits: analog 480-line video looks soft at 27 inch (it reads as arcade, not HD), and cheap upscalers add 1–3 frames. Gate item 7a measures glass-to-glass latency; over 120 ms and the 7-inch goes back on the cockpit with the 27-inch as spectator screen – nothing is wasted either way. Side benefit: the monitor lane now also has an independent marshal screen. Digital FPV (HDZero / Walksnail) would fix sharpness but costs Rs 15–30k per car plus receivers and reopens compliance – not for v1.

#	Error found in this run	Fix
6.1	7-inch was wired as the customer display on the monitor lane; no independent marshal view there	7-inch = marshal screen on both lanes
6.2	27-inch on the station 12 V SMPS would exceed its 5 A budget	27-inch on its own mains adaptor
6.3	Video pack had no low-voltage warning – video would just die mid-session	LiPo buzzer alarm on every pack; swap schedule every 4th car swap
6.4	Spares kit had no video pack or upscaler	2 packs + 1 upscaler added
6.5	Gate had no display-latency test	Item 7a added (15-point gate)

Budget after run 6: monitor station Rs 1.32–1.62 L, goggles station Rs 1.37–1.78 L, two stations Rs 2.69–3.40 L, programme Rs 3.1–4.9 L.

What did not change

Control chain topology (G29 → Pi → ELRS → ER4), 3-layer stop concept, speed caps, camera cartridge design, 25 mW Raceband inside 5725–5875 MHz, compliance stance (WPC/ETA query before bulk purchase), fleet-model-first ordering rule, prototype-before-duplicate.

Still to prove on the bench (marked verify in the plan)

- USB-UART at 400 k baud into the module's half-duplex CRSF pin (CP2102N / FT232RL).
- ELRS TX module behaviour when CRSF input stops — whether it stops RF on its own or the watchdog is doing all the work.
- 5 V 2 A UBEC and 2S 450–1000 mAh video packs: parts and prices (not verified live this pass).
- CVBS-to-HDMI upscaler latency on the 27-inch (gate item 7a).
- SpeedyBee TX800 25 mW output and pit mode on the bench VTX.